AI SKILL GAP INTELLIGENCE

Frontend Vision & Product Design Idea

Goal: Build a premium AI career-intelligence web application — not a basic Streamlit-style dashboard. It should feel like a real AI SaaS product.

1. Product Concept

The frontend is centered around **Career Intelligence**. A student uploads a resume and selects a target role. The system analyzes the resume, extracts skills, compares them with job requirements, identifies skill gaps, and presents job recommendations in a visual dashboard.

Signature concept: Career Genome — the student's current skills form a visual career profile that connects toward the missing skills required for the target career.

2. Technology Stack

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React.js	Main frontend application
Vite	Fast React development/build setup
Tailwind CSS	Modern responsive UI
Framer Motion	Premium animations/transitions
Recharts	Skill-gap charts and analytics
Lucide React	Clean icons
Axios	Flask API communication

3. Visual Theme

Futuristic and AI-inspired, but professional. Avoid excessive neon or a generic college-dashboard appearance.

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Role**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=10, greek=0, italic=0, link=[], rise=0, text='Role', textColor=Color(.2,.254902,.333333,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Color**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=10, greek=0, italic=0, link=[], rise=0, text='Color', textColor=Color(.2,.254902,.333333,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Primary / Midnight	#0B1020
Indigo	#6366F1
Cyan Accent	#22D3EE
Soft Violet	#A78BFA
Light Background	#F8FAFC
Main Text	#0F172A

4. Three-Page Application

PAGE 1 — Landing / Home

Purpose: Create the premium first impression and clearly explain the product.

Hero:
Discover the skills your career needs.
Upload your resume. Let AI understand your skills, identify your gaps, and guide your next career move.

CTA: Analyze My Resume →
Badges: NLP • AI • Career Intelligence

Visual: subtle animated skill nodes and connecting lines in the background; nodes slowly move/pulse.

Feature cards: NLP Resume Understanding • Skill Gap Analysis • Career Recommendations

PAGE 2 — Resume Analyzer

Primary interaction page: resume upload + target role selection + analysis.

Components: upload/drop zone • file info • target-role selector/input • Analyze Resume button • validation/error messages.

Loading experience:

```
✓ Resume processed
✓ Skills extracted
■ Comparing with job requirements
■ Building career profile
```

PAGE 3 — Career Intelligence Dashboard

The **showstopper page**. Turn NLP/ML output into an understandable visual story.

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Career Match	Animated match percentage + target role
Skill DNA	Current skills as chips, bars, radar/network visualization
Skill Gap	Missing skills with priority/severity
Job Recommendations	Recommended roles with match percentages
Career Path	Current profile → target role → skills to develop

5. Signature Feature — Career DNA

Do not only list skills. Visually represent the student's **Career DNA**: current skills form a profile that connects toward the selected target role.

```
Python ■■ SQL ■■ Excel ■■ Power BI
              ■
        Current Profile
             ↓
        DATA ANALYST
             ↓
   Advanced SQL • Tableau • Statistics
```

This becomes the visual identity of the project and differentiates it from a normal resume analyzer.

6. Animation Strategy

• Hero: fade/slide-in text and subtle skill-network motion.

• Cards: gentle hover lift/scale and soft shadow transitions.

• Analysis: animated NLP → matching → analysis pipeline.

• Match score: circular/progress animation from 0 to the calculated percentage.

• Skill gap: progress bars animate into their final values.

• Dashboard: sections reveal smoothly.

• Page transitions: subtle Framer Motion transitions.

7. Navigation

Home Analyze Intelligence

Keep the application focused: a polished 3-page experience rather than many unfinished screens.

8. Team Work Split

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Member**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=10, greek=0, italic=0, link=[], rise=0, text='Member', textColor=Color(.2,.254902,.333333,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Primary responsibility**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=10, greek=0, italic=0, link=[], rise=0, text='Primary responsibility', textColor=Color(.2,.254902,.333333,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

You	Frontend functionality, React setup, resume upload, API integration, result rendering, loading/error states, skill/job data integration
Friend	UI/UX, landing page, visual design, dashboard layout, cards, animations, charts, responsive polish

9. Complete Architecture

```
React + Vite + Tailwind + Framer Motion + Recharts + Lucide + Axios
↓
Flask API
↓
NLP Resume / Skill Extraction
↓
ML Skill Matching / Gap Analysis / Recommendations
↓
PostgreSQL
```

10. Design Principle

> **“It should look like a real AI career product, not a college dashboard.”**

Clean, premium, responsive, understandable, and visually connected to the AI/NLP/ML purpose. Animations should support the story rather than distract from the data.